Risk Assessment & Mitigation

Home Audio System — EE22005 Project Week 4 | Group HAS (BTA26, FS881, AC3847)

Severity (a)	Likelihood of Occurrence (b)
1 – Trivial < 30 min delay; no milestone affected; absorbed within normal task variation.	**1 – Remote** < 5 % probability of occurring during PW4 (≈ once per 20 project weeks).
2 – Minor 30 min – 2 h delay; absorbed by a buffer task; no milestone slips.	**2 – Unlikely** 5 – 20 % probability (≈ once per 5–10 project weeks).
3 – Moderate 2 – 4 h delay; one milestone slips by ≤ half a day; rework required.	**3 – Possible** 20 – 50 % probability (roughly even chance during this project week).
4 – Serious 4 – 8 h delay; critical-path milestone missed; a subsystem remains unverified at end of day.	**4 – Likely** 50 – 80 % probability (more likely than not to occur during PW4).
5 – Catastrophic > 8 h delay or safety incident; project week deadline missed; system cannot be demonstrated.	**5 – Very likely** > 80 % probability (expected to occur at least once during PW4).

Risk Assessment Matrix

(B) ↓ (A) →	Trivial	Minor	Moderate	Serious	Catastrophic
Remote	1	2	3	4	5
Unlikely	2	4	6	8	10
Possible	3	6	9	12	15
Likely	4	8	12	16	20
Very likely	5	10	15	20	25

Risk Rating Bands (A × B)

LOW RISK (1 – 8)	MEDIUM RISK (9 – 12)	HIGH RISK (15 – 25)
Continue, but review periodically to ensure controls remain effective.	Continue, but implement additional reasonably practicable controls where possible and monitor regularly.	STOP THE ACTIVITY Identify new controls. Activity must not proceed until risks are reduced to a low or medium level.

#	Hazard(s) identified	Existing controls & measures	Severity (a)	Likelihood (b)	Risk Rating (a × b)	Additional control / action required (20–50 words each)	Sev erit y (a)	Likelihood (b)	Risk Rating (a × b)
1	Class D half-bridge shoot-through — insufficient dead time between gate drives causes simultaneous conduction, destroying MOSFETs and halting WP3.	Sequential verification procedure: gate-drive signals measured on oscilloscope before connecting the load. RC dead-time network included in design.	4	4	16	Increase RC dead-time constant using Self's published values before first power-up. Add a hard hold-point in Test Protocol 2 requiring ≥ 0.5 μs dead time on both transitions before proceeding.	4	1	4
2	DC offset at RLC filter output exceeds 0.4 V — damages earphone transducer or causes hearing discomfort when earphones are connected.	DC blocking capacitor in series with earphone return. Oscilloscope DC-coupled check planned before earphone connection.	5	3	15	Formalise hold-point: no earphone connection until DC offset confirmed < 0.4 V under three signal conditions (no signal, low, full) and signed off by the test lead (FS881).	5	1	5
3	Integration impedance mismatch between equalizer output and Class D input causes signal loss > 3 dB at connection, delaying M4c past end of Day 5.	Subsystems built and tested individually before integration (sequential verification).	4	3	12	Measure EQ output impedance and Class D input impedance before connecting. Reserve WP4 buffer (2 h) specifically for integration debugging. Add a unity-gain buffer stage if mismatch exceeds 10:1.	4	2	8
4	Team member absent for 1+ days due to illness or interview — reduces capacity on critical-path tasks and risks milestone slippage.	Daily 09:00 stand-up to flag attendance. Team charter defines backup coverage for each role.	4	3	12	Pair-work all critical-path tasks so that a second team member can continue immediately. Reassign tasks at next stand-up if absence confirmed. Expand buffer if needed.	4	2	8
5	Insufficient time for final integration and QMS sign-off — subsystem delays cascade and M5 (file index complete) is missed at end of Day 5.	Gantt chart with critical path highlighted and buffer tasks after each work package.	5	2	10	Daily stand-up reviews Gantt baseline vs actual. If any WP buffer is fully consumed, de-scope non-critical features (e.g. spectrogram animation) to protect integration time.	5	1	5

6	BPF centre frequencies deviate beyond ±15 % of target after E12 component rounding, requiring redesign of one or more filter stages.	Component values pre-calculated in LTspice with E12 rounding before any build. Sweep each BPF individually before summing.	3	3	9	Run LTspice simulation with actual E12 values and confirm deviation < 15 % before building. Keep spare E12 components at the bench for rapid substitution if first sweep fails.	3	1	3
7	Critical component (e.g. TL072 op-amp, specific E12 capacitor) unavailable from lab stock — blocks BPF or Class D build.	All designs use E12 standard values available in the EE Lab stock cupboard.	3	3	9	Stock check at start of Day 1 for all BOM items. Hold two spare TL072 op-amps and one spare of each critical capacitor value at the bench throughout the week.	3	2	6
8	Loose breadboard wire or cold joint causes intermittent fault during demonstration — undermines M4c verification and demo credibility.	Visual inspection of breadboard against schematic at each build stage.	2	4	8	Pre-power continuity check (Test Protocol 2 §6.2) on all critical nets. Gentle flex test of wiring before sign-off. Keep build physically undisturbed between final verification and demo.	2	2	4